

Convex Analysis and Monotone Operator Theory in Hilbert Spaces

Chapter 13: Conjugation

Based on Bauschke & Combettes

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Why conjugation?

Big picture. Throughout Chapters 8–12 we described a convex function f by its *values* $f(x)$ at *points* x . Conjugation gives us a completely different, and often more powerful, description: we describe f by how it compares to *every linear functional* $\langle \cdot \mid u \rangle$.

The one-sentence idea

$f^*(y)$ measures the largest gap between the linear function $\langle y \mid \cdot \rangle$ and f — it encodes f from the “slope” perspective instead of the “point” perspective.

This is the continuous analogue of describing a convex polygon either by its vertices (points) or by its edges/supporting half-planes (slopes). Conjugation is the single most important construction in the theory: it underlies duality (Ch. 15), subdifferentials (Ch. 16), and proximal algorithms used throughout convex optimization and monotone operator splitting.

Notation you'll need

- $\mathcal{H}, \mathcal{K}$: real Hilbert spaces
- $\langle x \mid y \rangle$: inner product; $\|x\|$: induced norm
- $f : \mathcal{H} \rightarrow [-\infty, +\infty]$: extended-real-valued function
- $\text{dom } f = \{x : f(x) < +\infty\}$
- $\text{epi } f = \{(x, \xi) \in \mathcal{H} \times \mathbb{R} : f(x) \leq \xi\}$: epigraph
- f *proper*: $f \not\equiv +\infty$ and $f > -\infty$
- $\Gamma(\mathcal{H})$: lower semicontinuous (lsc) convex functions $\mathcal{H} \rightarrow]-\infty, +\infty]$
- $\Gamma_0(\mathcal{H})$: *proper* members of $\Gamma(\mathcal{H})$
- ι_C : indicator function of C (0 on C , $+\infty$ off C)
- $\sigma_C = (\iota_C)^*$: support function of C
- $q = \frac{1}{2}\|\cdot\|^2$
- $f \square g$: infimal convolution,
 $(f \square g)(x) = \inf_y (f(y) + g(x - y))$
- $\bar{f}$: lsc hull (closure) of f ; $\check{f}$: convex lsc hull of f
- $B(0; \rho)$: closed ball of radius ρ about 0
- $K^\ominus$: polar cone of K ; $V^\perp$: orthogonal complement
- $\mathcal{B}(\mathcal{H}, \mathcal{K})$: bounded linear operators $\mathcal{H} \rightarrow \mathcal{K}$

Definition 13.1: the Fenchel conjugate

Intuition. For a fixed “slope” (vector) $u \in \mathcal{H}$, compare the linear function $x \mapsto \langle x \mid u \rangle$ against f everywhere, and record the *largest* amount by which the linear function can exceed f . That largest gap, as a function of u , is the conjugate.

Definition (Fenchel conjugate)

Let $f : \mathcal{H} \rightarrow [-\infty, +\infty]$. The *conjugate* of f is the function $f^* : \mathcal{H} \rightarrow [-\infty, +\infty]$ defined by

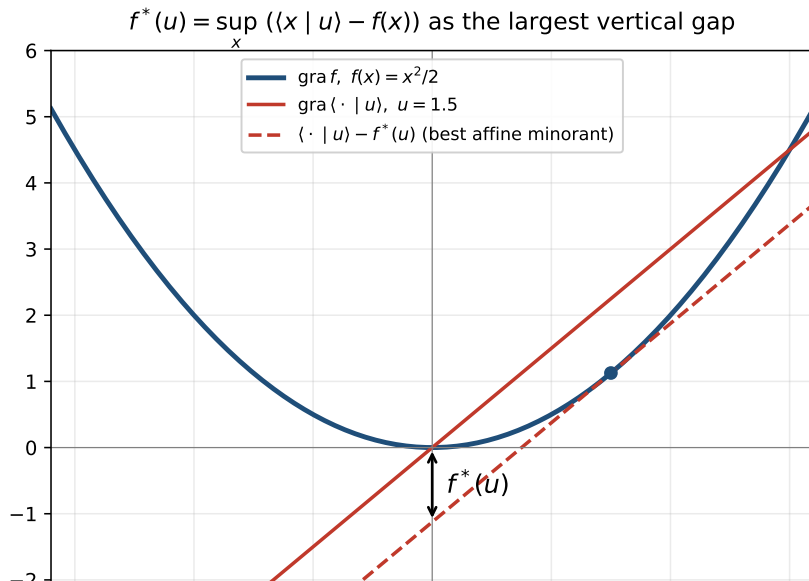
$$f^*(u) = \sup_{x \in \mathcal{H}} (\langle x \mid u \rangle - f(x)). \quad (13.1)$$

Here u ranges over $\mathcal{H}$ (thought of as the space of “slopes”), and the supremum is over all points x where we test f against the linear functional $\langle \cdot \mid u \rangle$.

Equivalently (Proposition 13.10(iv)): restricting the supremum to $\text{dom } f$,

$$f^*(u) = \sup_{x \in \text{dom } f} (\langle x \mid u \rangle - f(x)) = \sup_{(x, \xi) \in \text{epi } f} (\langle x \mid u \rangle - \xi).$$

Geometric picture (Fig. 13.1)



Derivation 1/2: conjugate of $f(x) = x^2/2$ (self-duality)

Setup. $\mathcal{H} = \mathbb{R}$, $f(x) = x^2/2 = q(x)$.

Step 1 (write the sup).

$$f^*(u) = \sup_{x \in \mathbb{R}} \left(ux - \frac{1}{2}x^2 \right).$$

Step 2 (optimize over x). The map $x \mapsto ux - \frac{1}{2}x^2$ is concave and differentiable; set the derivative to zero:

$$\frac{d}{dx} \left(ux - \frac{1}{2}x^2 \right) = u - x = 0 \implies x^* = u.$$

Step 3 (plug back in).

$$f^*(u) = u \cdot u - \frac{1}{2}u^2 = \frac{1}{2}u^2.$$

Conclusion: self-conjugacy

$f^*(u) = \frac{1}{2}u^2 = f(u)$ for every u , i.e. $f^* = f$. This is Example 13.6 and is a special case of Proposition 13.19: among all functions on $\mathcal{H}$, $q = \frac{1}{2}\|\cdot\|^2$ is the *only* one equal to its own conjugate.

Derivation 2/2: conjugate of $f(x) = |x|$

Setup. $\mathcal{H} = \mathbb{R}$, $f(x) = |x|$.

Step 1 (write the sup).

$$f^*(u) = \sup_{x \in \mathbb{R}} (ux - |x|).$$

Step 2 (split by sign of x). For $x > 0$: $ux - x = x(u - 1)$; for $x < 0$: $ux + x = x(u + 1)$. Each is linear in x , so:

- If $|u| \leq 1$: both slopes $u - 1 \leq 0$ and $u + 1 \geq 0$ push the supremum to $x = 0$, giving value 0.
- If $u > 1$: letting $x \rightarrow +\infty$ makes $x(u - 1) \rightarrow +\infty$.
- If $u < -1$: letting $x \rightarrow -\infty$ makes $x(u + 1) \rightarrow +\infty$ (since $u + 1 < 0$ and $x \rightarrow -\infty$).

Step 3 (assemble).

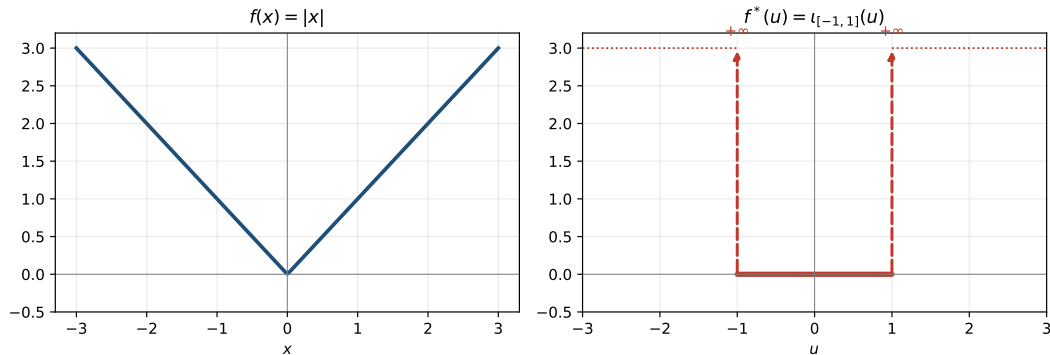
$$f^*(u) = \begin{cases} 0, & |u| \leq 1, \\ +\infty, & |u| > 1. \end{cases} = \iota_{[-1,1]}(u).$$

Conclusion

$\|\cdot\|^* = \iota_{[-1,1]}$: the conjugate of the “kinked” norm on $\mathbb{R}$ is the indicator of the interval $[-1, 1]$ (matches Example 13.3(v) in $\mathcal{H} = \mathbb{R}$, where $\|\cdot\|^* = \iota_{B(0;1)}$).

Figure: $f = |\cdot|$ and its conjugate side by side

Fenchel conjugation turns a "kink" into a "hard wall": $f = |\cdot| \leftrightarrow f^* = \iota_{[-1,1]}$



Conjugation converts the single kink of $|x|$ at 0 into a “hard wall” at $u = \pm 1$: any slope with $|u| > 1$ can beat f by an unbounded amount, while any slope with $|u| \leq 1$ can never beat it at all.

Running numeric example

$f(x) = x^2/2$, **using** $f^*(u) = u^2/2$:

u	0	1	2	-3
$f^*(u) = u^2/2$	0	0.5	2	4.5

$f(x) = |x|$, **using** $f^*(u) = \iota_{[-1,1]}(u)$:

u	0	0.5	1	1.5	-2
$f^*(u)$	0	0	0	$+\infty$	$+\infty$

Check Fenchel–Young (Proposition 13.15) at $x = 1$, $u = 2$ for $f = |\cdot|$:

$f(1) + f^*(2) = 1 + \infty \geq \langle 1 \mid 2 \rangle = 2$. Equality would require $x = 1$ to attain the sup defining $f^*(2)$; since $f^*(2) = +\infty$ and $f(1) = 1$ is finite, the inequality is strict but trivially true.

Example 13.2: a gallery of conjugates (I)

Intuition. Many named functions in information theory and statistics (negative entropies, log-barriers) are conjugates of elementary functions – this is why conjugation shows up throughout optimization and statistical estimation.

Selected pairs (f, f^*) on $\mathbb{R}$

- $f(x) = 1/x$ on $\mathbb{R}_{++}$, $+\infty$ else $\implies f^*(u) = -2\sqrt{-u}$ for $u \leq 0$, $+\infty$ for $u > 0$.
- $f(x) = -\ln(x)$ on $\mathbb{R}_{++}$ (negative Burg entropy) $\implies f^*(u) = -\ln(-u) - 1$ for $u < 0$, $+\infty$ for $u \geq 0$.
- $\cosh^*(u) = u \operatorname{arcsinh}(u) - \sqrt{u^2 + 1}$.
- $\exp^*(u) = u \ln(u) - u$ for $u > 0$; 0 at $u = 0$; $+\infty$ for $u < 0$ (negative Boltzmann–Shannon entropy).

Example 13.2: a gallery of conjugates (II)

More entropy-type pairs

- Negative Fermi–Dirac entropy $f(x) = x \ln x + (1 - x) \ln(1 - x)$ on $]0, 1[$ (with $f = 0$ at $\{0, 1\}$, $+\infty$ else) has $f^*(u) = \ln(1 + e^u)$; the “flip” $(f^*)^\vee$ is the *logistic loss*.
- Negative Bose–Einstein entropy $f(x) = x \ln x - (x + 1) \ln(x + 1)$ on $\mathbb{R}_{++}$ has

$$f^*(u) = \begin{cases} -\ln(1 - e^u), & u < 0, \\ +\infty, & u \geq 0. \end{cases}$$

- $f(x) = \sqrt{1 + x^2}$ has $f^*(u) = -\sqrt{1 - u^2}$ for $|u| \leq 1$, $+\infty$ for $|u| > 1$.

These all follow the same recipe as our two derivations: write the sup, differentiate (or bound), solve, and check the domain of validity.

Example 13.3: conjugates of sets and cones

Intuition. Indicator functions convert set geometry into function language; their conjugates are exactly the classical support functions and polars from Chapter 7.

Direct applications of (13.1)

- $f = \iota_C$ for $C \subset \mathcal{H} \implies f^* = \sigma_C$ (support function of C).
- $f = \iota_K$, K a nonempty cone $\implies f^* = \sigma_K = \iota_{K^\ominus}$ ($K^\ominus$: polar cone).
- $f = \iota_V$, V a linear subspace $\implies f^* = \iota_{V^\ominus} = \iota_{V^\perp}$.
- $f = \iota_{B(0;1)} \implies f^* = \sigma_{B(0;1)} = \|\cdot\|$.
- $f = \|\cdot\| \implies f^* = \iota_{B(0;1)}$ (the $\mathcal{H} = \mathbb{R}$ instance of this is exactly our derivation of $|\cdot|^*$ above!).

Examples 13.4–13.9: quadratic perturbations

Intuition. Adding a scaled quadratic $\gamma^{-1}q$ to φ before conjugating produces a clean transformation rule — this is the seed of the *Moreau envelope* machinery used later in the book.

Quadratic perturbation formula (Example 13.4)

Let φ be proper, $\gamma \in \mathbb{R}_{++}$, $f = \varphi + \gamma^{-1}q$. Then

$$f^* = \gamma q - \gamma \varphi \circ \gamma \text{Id} = \gamma q - (\varphi \square (\gamma^{-1}q)) \circ \gamma \text{Id}, \quad (13.4)$$

where $\gamma \varphi = \gamma \varphi(\cdot/\gamma)$ is the *epi-multiple* of φ by γ .

Consequences.

- $f = \iota_C + q/2 \Rightarrow f^* = (\|\cdot\|^2 - d_C^2)/2$ (Example 13.5); with $C = \mathcal{H}$ this again gives self-conjugacy of q .
- $f = \iota_{B(0;\rho)} + q/2$: $f^*(u) = \rho\|u\| - \rho^2/2$ for $\|u\| > \rho$, $\|u\|^2/2$ for $\|u\| \leq \rho$ — the *Huber function* in $\mathbb{R}$ (Example 13.7).

Proposition 13.10: first consequences of the definition

Intuition. Before deriving anything deep, record the “free” facts that fall directly out of (13.1).

Basic facts

- (i) $f^*(0) = -\inf f(\mathcal{H})$.
- (ii) $-\infty \in f^*(\mathcal{H}) \iff f \equiv +\infty \iff f^* \equiv -\infty$.
- (iii) If f^* is proper, then f is proper.
- (iv) $f^*(u) = \sup_{x \in \text{dom } f} (\langle x \mid u \rangle - f(x)) = \sup_{(x, \xi) \in \text{epi } f} (\langle x \mid u \rangle - \xi)$.
- (v) $f^* = \iota_{\text{epi } f}^*(\cdot, -1) = \sigma_{\text{epi } f}(\cdot, -1)$.
- (vi) If f is proper, $f^* = \iota_{\text{gra } f}^*(\cdot, -1) = \sigma_{\text{gra } f}(\cdot, -1)$.

Item (i) already tells you $f^*(0)$ for free: e.g. for $f(x) = x^2/2$, $f^*(0) = -\inf f(\mathbb{R}) = 0 = f(0)$, consistent with self-conjugacy.

Affine minorants $\leftrightarrow$ points of $\text{epi } f^*$

Proposition 13.12

Let $f : \mathcal{H} \rightarrow [-\infty, +\infty]$. Then:

- (i) $(u, \mu) \in \text{epi } f^* \iff \langle \cdot \mid u \rangle - \mu \leq f$.
- (ii) $f^* \equiv +\infty$ iff f has no continuous affine minorant.
- (iii) If $\text{dom } f^* \neq \emptyset$, then f is bounded below on every bounded subset of $\mathcal{H}$.

Intuition. This is the precise bridge between the “slope” picture and the “point” picture: every point (u, μ) under the graph of f^* corresponds *exactly* to a continuous affine function $\langle \cdot \mid u \rangle - \mu$ lying entirely below f .

Proposition 13.13: the conjugate is always nice

Key structural fact

For any $f : \mathcal{H} \rightarrow [-\infty, +\infty]$ (not assumed convex or lsc!), the conjugate satisfies

$$f^* \in \Gamma(\mathcal{H}),$$

i.e. f^* is always convex and lower semicontinuous.

Why. By Proposition 13.10(iv), f^* is a pointwise supremum of the affine (hence lsc convex) functions $(\langle x \mid \cdot \rangle - \xi)_{(x, \xi) \in \text{epi } f}$, and a supremum of lsc convex functions is lsc and convex (Proposition 9.3). Conjugation is a “convexifying, lower-semicontinuizing” operation — no matter how wild f is.

Proposition 13.15: the Fenchel–Young inequality

Intuition. Directly from the definition of a supremum, $\langle x \mid u \rangle - f(x) \leq f^*(u)$ for every pair (x, u) — rearranged, this is the single most-used inequality in convex analysis.

Fenchel–Young inequality

Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ be proper. Then

$$(\forall x \in \mathcal{H})(\forall u \in \mathcal{H}) \quad f(x) + f^*(u) \geq \langle x \mid u \rangle. \quad (13.10)$$

- $f(x)$: value of the primal function at x ;
- $f^*(u)$: value of the conjugate at u ;
- $\langle x \mid u \rangle$: the “coupling” between the two spaces.

Numeric check. For $f = q$, $x = u = 2$: $f(2) + f^*(2) = 2 + 2 = 4 = \langle 2 \mid 2 \rangle$: *equality* holds (self-conjugate case, always tight at $x^* = u$). For $f = |\cdot|$, $x = 2$, $u = 2$: $f(2) + f^*(2) = 2 + \infty \geq 4$: inequality only, since $u = 2 \notin [-1, 1]$.

Proposition 13.16: biconjugation

Intuition. Conjugating twice can only make the function smaller or equal (a “regularization”); conjugating three times is the same as conjugating once.

Biconjugate facts

- (i) $f^{**} \leq f$ always.
- (ii) $f \leq g \Rightarrow f^* \geq g^*$ and $f^{**} \leq g^{**}$ (conjugation reverses order).
- (iii) $f^{***} = f^*$.
- (iv) $(\check{f})^* = f^*$ ($\check{f}$: convex lsc hull of f).

Warning (Example 13.17). These inequalities can be extreme: for $f(x) = -|x|$ on $\mathbb{R}$, the convex hull $\check{f} \equiv -\infty$ and $f^* \equiv +\infty$ everywhere — conjugation of a concave-looking function destroys all information.

Proposition 13.19: Self-conjugacy is rigid

Self-conjugacy

$$f = f^* \iff f = \frac{1}{2} \|\cdot\|^2 = q.$$

Proof sketch. ($\Leftarrow$) is Example 13.6. ($\Rightarrow$): if $f = f^*$ then f is proper (Prop. 13.10(ii)), and Fenchel–Young gives $2f(x) \geq \langle x \mid x \rangle$, i.e. $f \geq q$; applying (13.16)(ii) to $f \geq q$ gives $f^* \leq q^* = q$, i.e. $q \geq f^* = f \geq q$, so $f = q$.

Consequence (Remark 13.20). A convex cone $K \neq \mathcal{H}$ can never be self-polar in the strong sense $\iota_K = \iota_K^*$, because that would force $\iota_K = q$, impossible for an indicator function. In particular a linear subspace can never be self-orthogonal.

This is exactly why our two running examples are so instructive: q is the *unique* fixed point of conjugation, while $|\cdot|$ is very much *not* one ($|\cdot|^* = \iota_{[-1,1]} \neq |\cdot|$).

Symmetry and scaling rules (Props. 13.21–13.23)

Useful calculus rules

- If f is even, then f^* is even (Prop. 13.21).
- If $f \geq f(0) = 0$, then $f^* \geq f^*(0) = 0$ (Prop. 13.22).
- For $\alpha \in \mathbb{R}_{++}$: $(\alpha f)^* = \alpha f^*(\cdot/\alpha)$ and $(\alpha f(\cdot/\alpha))^* = \alpha f^*$ (Prop. 13.23(i)-(ii)).
- Translation: $(\tau_y f + \langle \cdot \mid v \rangle + \beta)^* = \tau_v f^* + \langle y \mid \cdot \rangle - \langle y \mid v \rangle - \beta$ (Prop. 13.23(iii)).
- Bijective $L \in \mathcal{B}(\mathcal{H})$: $(f \circ L)^* = f^* \circ L^{*-1}$ (Prop. 13.23(iv)).

Sanity check with our example. $f = |\cdot|$, $\alpha = 2$: $(2|\cdot|)^*(u) = 2\iota_{[-1,1]}(u/2) = 2\iota_{[-2,2]}(u) = \iota_{[-2,2]}(u)$ (indicator is $\{0, \infty\}$ -valued, so the “ $2\times$ ” in front does not change the values — only the set changed, from $[-1, 1]$ to $[-2, 2]$, exactly as scaling the slope threshold predicts.

Proposition 13.24: infimal convolution $\leftrightarrow$ sum

Intuition. This is the single most important calculus rule for conjugates: it says that “inf-convolving” in the primal space corresponds *exactly* to adding in the conjugate (dual) space.

Key duality formula

$$(f \square g)^* = f^* + g^*. \quad (13.17)$$

Here $f \square g$ is the infimal convolution $(f \square g)(x) = \inf_{y \in \mathcal{H}} (f(y) + g(x - y))$: it is the function-space analogue of Minkowski-summing epigraphs. If f, g are proper, then also $(f + g)^* \leq f^* \square g^*$ (Prop. 13.24(ii)) — equality generally requires a qualification condition (developed in Ch. 15).

Also in Prop. 13.24: $(\gamma f)^* = f^* + (\gamma/2)\|\cdot\|^2$; for $L \in \mathcal{B}(\mathcal{H}, \mathcal{K})$, $(L \triangleright f)^* = f^* \circ L^*$ (image/epi-composition rule) and $(f \circ L)^* \leq L^* \triangleright f^*$.

Propositions 13.28–13.30: families and direct sums

Infima/suprema of families

For a family $(f_i)_{i \in I}$ of proper functions:

- $(\inf_{i \in I} f_i)^* = \sup_{i \in I} f_i^*$ (Prop. 13.28(i), *exact equality*).
- $(\sup_{i \in I} f_i)^* \leq \inf_{i \in I} f_i^*$ (Prop. 13.28(ii), only $\leq$ in general; with a sharpening to $\check{\cdot}$ once all $f_i \in \Gamma_0$, Prop. 13.47).

Separable sums (Prop. 13.30)

For a finite family of Hilbert spaces $\mathcal{H}_i$ and functions f_i ,

$$\left(\bigoplus_{i \in I} f_i \right)^* = \bigoplus_{i \in I} f_i^*. \quad (13.25)$$

Conjugation of a “block-separable” function is just the block-separable conjugation of each piece — e.g. this is exactly how Example 13.31 derives $((1/p)\|\cdot\|_p^p)^* = (1/p^*)\|\cdot\|_{p^*}^{p^*}$ ($1/p + 1/p^* = 1$) in $\mathbb{R}^N$, and Example 13.32 derives $\|\cdot\|_p^* = \iota_{B_{p^*}}$, i.e. the classical ℓ_p – ℓ_{p^*} duality via Hölder’s inequality.

Bivariate functions and autoconjugacy (13.33–13.36)

Intuition. Conjugation extends naturally to functions of two Hilbert-space variables; “marginalizing” one variable before or after conjugating commute in a controlled way.

Proposition 13.33

Let $F : \mathcal{H} \times \mathcal{K} \rightarrow]-\infty, +\infty]$ be proper and set $f(x) = \inf_{y \in \mathcal{K}} F(x, y)$. Then $f^* = F^*(\cdot, 0)$: the conjugate of a marginal is the conjugate evaluated at the dual variable of the marginalized coordinate set to 0.

Definition 13.34 (autoconjugate)

$F : \mathcal{H} \times \mathcal{H} \rightarrow [-\infty, +\infty]$ is *autoconjugate* if $F^* = F^\top$, where $F^\top(u, x) = F(x, u)$ (swap the two arguments). Proposition 13.36: if $F \in \Gamma(\mathcal{H} \oplus \mathcal{H})$ is autoconjugate, then $F \geq \langle \cdot | \cdot \rangle$ and $F^* \geq \langle \cdot | \cdot \rangle$.

Theorem 13.37: the Fenchel–Moreau theorem

Intuition. We already know $f^{**} \leq f$ always (Prop. 13.16(i)), with equality failing whenever f is not itself convex and lsc. The Fenchel–Moreau theorem says this is the *only* obstruction: convexity + lower semicontinuity is exactly the property characterized by $f = f^{**}$.

Theorem (Fenchel–Moreau)

Let $f : \mathcal{H} \rightarrow]-\infty, +\infty]$ be proper. Then

$$f \text{ is lsc and convex} \iff f = f^{**}.$$

In this case, f^* is proper as well.

Why this matters. It means $\Gamma_0(\mathcal{H})$ (proper lsc convex functions) is *exactly* the set of functions that conjugation perfectly “round-trips” on. Conjugation is a genuine involution on $\Gamma_0(\mathcal{H})$, i.e. a duality — not just a one-way transform.

Proof sketch of Fenchel–Moreau

($\Leftarrow$) If $f = f^{**}$, then f is a conjugate function, hence lsc and convex by Proposition 13.13. This direction is immediate.

($\Rightarrow$), **main idea.** Suppose f is lsc and convex. Fix $x \in \mathcal{H}$ and $\xi < f(x)$; project (x, ξ) onto the closed convex set $\text{epi } f$ to get (p, π) . The projection characterization (Proposition 9.18) forces $\pi \geq \xi$ and a separating-hyperplane-type inequality. Two cases:

- If $\pi > \xi$: setting $v = (x - p)/(\pi - \xi)$, one shows $f^*(v) \leq \langle x | v \rangle - \pi$, hence $f^{**}(x) \geq \pi > \xi$.
- If $\pi = \xi$: a limiting argument using $w \in \text{dom } f^*$ and $\lambda \rightarrow +\infty$ shows $f^{**}(x) = +\infty = f(x)$ when $x \notin \text{dom } f$, or $f^{**}(x) = f(x)$ when $x \in \text{dom } f$.

Since $\xi < f(x)$ was arbitrary, $f^{**}(x) \geq f(x)$; combined with $f^{**} \leq f$ always, we get $f^{**} = f$.

Corollaries 13.38–13.42

Immediate consequences

- **Corollary 13.38.** $f \in \Gamma_0(\mathcal{H}) \Rightarrow f^* \in \Gamma_0(\mathcal{H})$ and $f^{**} = f$: conjugation is a bijection (in fact an involution) on $\Gamma_0(\mathcal{H})$.
- **Corollary 13.39.** For $f \in \Gamma_0(\mathcal{H})$: $f \leq g \iff f^* \geq g^*$ (order-reversing bijection).
- **Corollary 13.40.** For $f, g \in \Gamma_0(\mathcal{H})$: $f = g^* \iff g = f^*$.
- **Corollary 13.42.** Every $f \in \Gamma_0(\mathcal{H})$ is the *supremum of its continuous affine minorants* – the continuous, affine version of “every convex set is the intersection of its supporting half-spaces.”

Example 13.41. Combining Example 13.32 with Corollary 13.40 shows that the dual norm $\|\cdot\|_{p,*}$ on $\mathbb{R}^N$ equals $\|\cdot\|_{p^*}$: exactly the classical duality between ℓ_p and ℓ_{p^*} norms.

Example 13.43: recovering old friends

Applying Fenchel–Moreau to specific indicator functions recovers familiar closure/polarity identities:

Consequences of $(\iota_C)^{**} = \iota_{\overline{\text{conv}} C}$

- $C \subset \mathcal{H}$ nonempty: $\iota_C^{**} = \iota_{\overline{\text{conv}} C}$.
- V a closed subspace: $\iota_V^{**} = \iota_V$, recovering $V^{\perp\perp} = V$.
- K a nonempty closed convex cone: $\iota_K^{**} = \iota_K$, recovering $K^{\ominus\ominus} = K$ (bipolar theorem, Corollary 6.34).

This shows the abstract Fenchel–Moreau machinery reproduces very concrete geometric facts you already know from Chapters 3, 6, and 7.

Propositions 13.44–13.46: lsc hulls

Intuition. What happens at a single point x if f is convex but *not* lsc? Biconjugation still does something sensible: it computes the lsc hull.

Key equivalences (Prop. 13.44)

For f convex with $f(x) \in \mathbb{R}$, the following are equivalent: (i) f is lsc at x ; (ii) $f^{**}(x) = \check{f}(x) = \bar{f}(x) = f(x)$; (iii) $f^{**}(x) = f(x)$.

Proposition 13.45 / 13.46

If f has a continuous affine minorant, then $f^{**} = \check{f}$ (the lsc convex hull); otherwise $f^{**} \equiv -\infty$. For f proper convex with a continuous affine minorant: $\text{dom } f \subset \text{dom } f^{**} \subset \overline{\text{dom } f}$, $\text{epi } f^{**} = \overline{\text{epi } f}$, and $f^{**}(x) = \lim_{y \rightarrow x} f(y)$.

Propositions 13.49–13.51: recession & integral functionals

Conjugates and recession functions (Prop. 13.49)

For $f \in \Gamma_0(\mathcal{H})$:

$$\text{rec}(f^*) = \sigma_{\text{dom } f} \quad \text{and} \quad \text{rec } f = \sigma_{\text{dom } f^*}.$$

The “directions of recession” of one function are governed by the support function of the *domain* of the other.

Conjugation of integral functionals (Prop. 13.50)

On $\mathcal{H} = L^2((\Omega, \mathcal{F}, \mu); H)$, if $f(x) = \int_{\Omega} \varphi(x(\omega)) \mu(d\omega)$ (an integral functional built from $\varphi \in \Gamma_0(H)$), then under mild conditions $f^*(u) = \int_{\Omega} \varphi^*(u(\omega)) \mu(d\omega)$: conjugation commutes with integration.

Example 13.51. For X a square-integrable r.v. and $f(X) = \frac{1}{p} E|X|^p$, we get $f^*(X) = \frac{1}{p^*} E|X|^{p^*}$: the L^p – L^{p^*} duality for random variables.

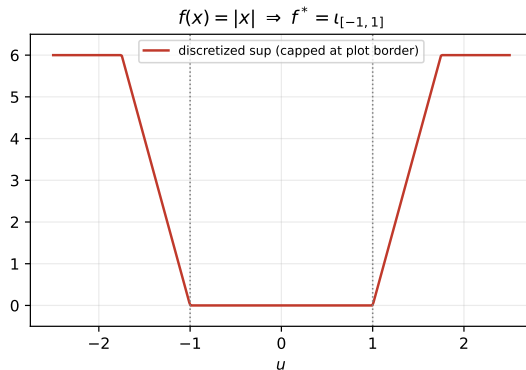
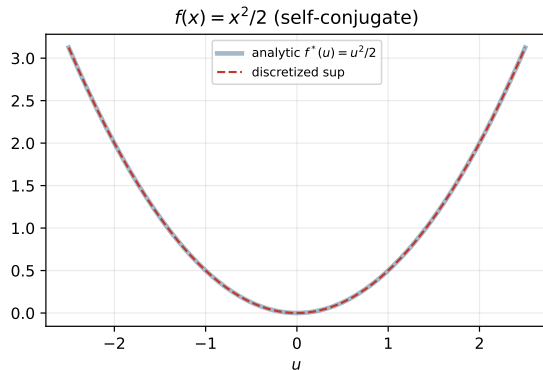
Python: conjugates via a discretized supremum

Idea. Since $f^*(u) = \sup_x (ux - f(x))$, we can *approximate* f^* on any finite grid of u -values by brute-force maximizing over a fine grid of x -values – no closed form needed. This is exactly how one would numerically explore conjugates of functions too complicated to differentiate by hand.

```
1 import numpy as np
2
3 def numerical_conjugate(f, x_grid, u_grid):
4     """f*(u) = sup_x (u*x - f(x)), brute force over a grid."""
5     fx = f(x_grid)                                # shape (len(x_grid),)
6     # outer(u,x) - f(x): shape (len(u_grid), len(x_grid))
7     vals = np.outer(u_grid, x_grid) - fx[None, :]
8     return np.max(vals, axis=1)
9
10 x_grid = np.linspace(-8, 8, 4001)
11 u_grid = np.linspace(-2.5, 2.5, 400)
12
13 f_quad = lambda x: 0.5 * x**2                    # self-conjugate
14 f_abs = lambda x: np.abs(x)                      # conjugate = indicator of [-1,1]
15
16 fstar_quad = numerical_conjugate(f_quad, x_grid, u_grid)
17 fstar_abs = numerical_conjugate(f_abs, x_grid, u_grid)
18
19 print(fstar_quad[u_grid == 1.5])                # ~1.125, matches u^2/2
20 print(fstar_abs[np.argmax(np.abs(u_grid-0.5))]) # ~0
21 print(fstar_abs[np.argmax(np.abs(u_grid-2.0))]) # large (approx +inf)
```

Numerical vs. analytic conjugates

Numerically evaluating the conjugate via a discretized supremum



The discretized supremum recovers $f^*(u) = u^2/2$ almost exactly for $f = q$ (a smooth, unconstrained optimization over x), while for $f = |\cdot|$ it correctly explodes as soon as $|u|$ exceeds 1, saturating at the largest value $\max(u \cdot x) - |x|$ achievable on the finite x -grid — a finite-grid stand-in for $+\infty$.

Summary: Chapter 13 in one slide

- **Definition.** $f^*(u) = \sup_x (\langle x | u \rangle - f(x))$: the largest gap between a linear functional and f .
- **Always true.** $f^* \in \Gamma(\mathcal{H})$ (convex, lsc), no matter what f is; Fenchel–Young: $f(x) + f^*(u) \geq \langle x | u \rangle$.
- **Self-duality is rigid.** $f = f^*$ forces $f = \frac{1}{2} \|\cdot\|^2$ (Prop. 13.19); e.g. $|\cdot|^* = \iota_{[-1,1]} \neq |\cdot|$.
- **Biconjugation.** $f^{**} \leq f$ always; *Fenchel–Moreau*: equality holds iff f is proper, lsc, and convex, and then f^* is proper too.
- **Calculus.** Inf-convolution $\leftrightarrow$ sum ($(f \square g)^* = f^* + g^*$); separable sums add coordinatewise; scaling/translation/linear-map rules all have clean conjugate analogues.
- **Where this leads.** Chapter 14 develops Moreau’s decomposition and further conjugation identities; Chapter 15 builds Fenchel–Rockafellar duality directly on these tools.